Poolin EOC07 Beginner's Guide: Your First Machine Embroidery

Oct 20, 2026 · @Joshua Henderson

How to use this guide

By the end of this guide you will have stitched a two-line name patch that reads **Henderson** over **est. 2007**. Each step is short: do it, check the result, then move on. Read a step, try it on the machine, and only come back to the text when something looks wrong.

- **Do it together.** One person reads the step out loud, the other does it. Swap at each step so you both learn every part.
- **Practice on scrap first.** Your first two or three runs should go on scrap cotton (an old pillowcase or dish towel). Mistakes cost nothing there.
- **Expect to re-thread.** Almost every beginner problem (nests of thread, skipped stitches, broken thread) is fixed by re-threading the top thread with the presser foot UP.
- **Keep your manual handy.** Poolin updates screens and menus between batches, so if a button name here differs slightly from your screen, trust your screen and the printed manual in the box.

Look for these markers along the way:

- **Part callout:** what a piece of the machine is and what it does.
- **Hand vs machine:** where your hand-stitching habits help or need adjusting.
- **Try it:** a quick hands-on experiment.
- **Check:** how to know a step worked before moving on.

Hand vs machine: what changes

Your hand skills transfer more than you think: you already know fabric, hoop tension, and color planning. The big change is that the machine uses **two threads** that lock together, and the **hoop moves while the needle stays still**.

Topic	By hand	On the EOC07
Threads	One strand, pulled all the way through	Two threads: **top thread** (from the spool) and **bobbin thread** (underneath). They lock in the middle of the fabric.
Thread type	Cotton floss, split into strands	40-weight polyester or rayon embroidery thread on spools. Never split it.
Back of the work	Neat back matters	The back will show white bobbin thread. That is normal and correct.
What moves	Your hand moves the needle	The needle only goes up and down. The **embroidery arm** slides the hoop around under it.
Pattern	Printed chart or transfer	A digital file (.DST) or text typed on the touchscreen
Behind the fabric	Usually nothing	**Stabilizer** every time. Without it the fabric puckers and stitches pull out of shape.
Hoop tension	Drum-tight, re-tighten as you go	Drum-tight, set once. You cannot touch it while stitching.
Color changes	Re-thread your needle	Machine stops; you swap the spool and re-thread the top thread. Bobbin stays white.
Speed	A few stitches a minute	Up to about 860 stitches per minute

Hand vs machine: the habit to break is pulling stitches to tighten them. On the machine, tension is set by dials and springs, not your fingers. If stitches look loose or tight, re-thread first, then adjust tension.

Meet the machine

The EOC07 is a single-needle embroidery machine with a 7-inch touchscreen and a largest stitch area of 7.9″ x 15″ (200 x 380 mm). Before plugging anything in, stand in front of it together and find each part below. Touch it, open it, close it.

Part	Where it is	What it does
Spool pin / thread stand	Top of the machine, or the separate stand in the box	Holds the spool of top thread.
Thread guides	Hooks and slots along the top and front	Lead the top thread down to the needle in a set path. Numbered or arrowed on most machines.
Tension disc	Inside the front slot the thread goes down	Squeezes the top thread so stitches lock correctly. The thread must sit *between* the discs.
Take-up lever	Metal arm with an eye that moves up and down at the front	Pulls up slack after every stitch. Missing it is the #1 cause of thread nests.
Needle and needle clamp	Bottom of the needle bar	Holds the needle. Flat side of the needle faces the back.
Needle threader	Small lever beside the needle	Pushes a tiny hook through the needle eye so you don't have to.
Presser foot (embroidery foot)	Small foot around the needle	Holds fabric down while the needle lifts. Must be **up** while threading, **down** while stitching.
Needle plate	Flat metal plate under the needle	Covers the bobbin area. The needle passes through its hole.
Bobbin and bobbin cover	Under a clear or metal cover in front of the needle	Holds the small bobbin of under-thread. The kit's bobbins are Class 15 (SA156).
Bobbin winder	Small post on top of the machine	Fills an empty bobbin from a spool.
Embroidery arm (carriage)	The flat moving unit at the back or side of the bed	Grabs the hoop and moves it around under the needle. Keep the area around it clear.

Part	Where it is	What it does
Hoop and hoop clamp	Clips onto the embroidery arm	Holds fabric drum-tight. Inner ring sits inside the outer ring; a screw tightens the outer ring.
Touchscreen	Front panel	Choose or type a design, resize, move it, and start or stop.
Start/Stop button	Near the needle, on the front	Starts and pauses stitching. Your emergency stop.
Thread cutter	Automatic, near the needle	Trims threads at color changes and at the end (if your model has it switched on).
USB port	Side of the machine	Load .DST files from a flash drive. Wi-Fi is the other way to send designs.
Power switch and cord socket	Right side or back	On/off.

Try it: with the machine unplugged, raise and lower the presser foot lever five times. Then turn the handwheel (the big knob on the right side) slowly toward you and watch the take-up lever and needle move together. Always turn it **toward you**, never away.

Supplies checklist

Gather these before Step 1. Most came in your box or bundle.

- [ ] **Top thread:** one dark spool for "Henderson" (navy, black, or forest green) and optionally a second color for "est. 2007". 40-weight polyester embroidery thread.
- [ ] **Bobbin thread:** pre-wound white bobbins (Class 15 / SA156). Bobbin thread is thinner than top thread; don't swap them.
- [ ] **Embroidery needles:** size 75/11 for cotton and most fabrics. Put in a fresh one for your first project.
- [ ] **Stabilizer:** medium-weight cut-away (in your kit) or tear-away for scrap practice. Cut a piece at least 1 inch larger than the hoop on every side.
- [ ] **Practice fabric:** plain woven cotton, about 12″ x 12″. Avoid stretchy T-shirt knit for your first try.

- [ ] **Hoop:** the 5.5″ x 5.5″ (140 x 140 mm) hoop for this project.
- [ ] **Small curved scissors or snips**, for trimming jump threads.
- [ ] **USB flash drive**, formatted FAT32, if you will load the pattern file.
- [ ] **Tweezers** (handy for threading and pulling stray threads).
- [ ] **Optional:** temporary spray adhesive, water-soluble topper for towels or fleece.

Hand vs machine: stabilizer is the one new material. It plays the role your hand and hoop played together: it stops the fabric from stretching while thousands of stitches pull on it. Cut-away stays on the back forever; tear-away rips off after.

Step 1: Set up the machine

Goal: machine on a sturdy table, powered on, showing the home screen, with room for the hoop to move. About 15 minutes.

1. **Pick the spot.** A solid table, not a folding one. The hoop travels up to 15 inches, so leave about 12 inches of clear space behind and beside the machine. Nothing should touch the embroidery arm.
2. **Unpack and remove all packing material.** Look for foam or tape around the embroidery arm and under the needle. Some machines have a shipping lock or a cardboard insert under the arm. Remove every piece.
3. **Set up the thread stand** (if your kit has a separate one) behind the machine.
4. **Plug in the power cord** to the machine, then the wall. Use a surge protector if you have one.
5. **Clear the arm, then switch on.** At power-on the embroidery arm moves to its home position. Keep hands and objects away from it.
6. **Look at the home screen.** You should see options for built-in designs, lettering or text, and USB or Wi-Fi files. Tap around. Nothing stitches until you press Start/Stop.
7. **Raise the presser foot** and leave it up for the next two steps.

Check: the arm moved and stopped, the screen is lit, and there is no packing material left. If the arm does not move or the screen shows an error, power off, re-check for packing pieces, and power on again.

Try it: find the settings or tools menu and look for screen brightness, language, and units. Switch units to inches if you prefer them.

Step 2: Load the bobbin

Goal: a white bobbin sitting in the bobbin area, with its thread caught in the tension spring. Skip winding for now: use a **pre-wound bobbin** from your kit.

1. **Power off** (any time your fingers go near the needle).
2. **Remove the hoop** if one is attached, and raise the needle by turning the handwheel toward you.
3. **Open the bobbin cover** in front of the needle (slide or lift it, depending on your cover).
4. **Drop the bobbin in** so the thread unwinds the way the diagram next to the bobbin area shows. On most drop-in systems the thread comes off **counter-clockwise**, making a backwards "P" when you look down at it.
5. **Guide the thread through the slot** and along the groove until it clicks into the tension spring. Leave a 4-inch tail.
6. **Close the cover.** You do not need to pull the bobbin thread up; the machine catches it on the first stitch.

Check: pull the tail gently. The bobbin should turn smoothly with slight resistance, not spin freely. If it spins freely, the thread missed the tension spring; take it out and redo step 5.

Hand vs machine: the bobbin is the "back half" of every stitch. You'll never see it on the front of a good stitch-out, but if it shows on top, your top thread is too tight or the bobbin is in backwards.

Later: winding your own bobbins. Put a spool on the pin, run the thread around the winder guide, wrap it a few times around an empty bobbin, push the bobbin onto the winder post, and start winding from the screen or by pushing the winder over. The machine stops when the bobbin is full. Use bobbin thread, not top thread.

Step 3: Thread the top thread

Goal: dark thread running from the spool, through every guide, through the take-up lever, and through the needle eye. This is the step you will repeat most, so do it twice: once each.

The golden rule: **presser foot UP while threading.** With the foot up, the tension discs open and the thread drops between them. With the foot down, the thread rides on top and you get a nest of loops under the fabric.

1. **Raise the presser foot** and turn the handwheel toward you until the take-up lever is at its **highest** point (most machines have a "needle up" button that does this).
2. **Place the spool** on the spool pin with a spool cap to stop it wobbling. Thread should come off the back or as your machine diagram shows.
3. **Follow the numbered guides** printed on the machine: across the top, into the first guide.
4. **Down the right channel** (through the tension disc), around the U-turn at the bottom, and **up the left channel**.
5. **Through the take-up lever eye** at the top, from right to left. Look for it: the thread must pass through the eye, not just across it.
6. **Down the left channel** to the needle bar guide just above the needle.
7. **Thread the needle** front to back. Use the needle threader: pull its lever down, hook the thread across the guide, release, and pull the loop through.
8. **Pull 4 inches of tail** through the needle and lay it under the presser foot, toward the back.

Check: with the presser foot **down**, pull the thread near the needle. You should feel firm, even resistance. Raise the foot and pull again: it should slide easily. That difference means the thread is in the tension discs. No difference? Re-thread from step 1.

Try it: each of you thread the machine once from scratch, while the other watches for a missed take-up lever.

Step 4: Hoop the fabric

Goal: fabric and stabilizer together, drum-tight in the 5.5" x 5.5" hoop, clipped onto the embroidery arm. This is where your hand-hooping experience pays off.

1. **Mark the center.** Fold your fabric in quarters and finger-press, or draw a small cross with a washable marker. The cross is where the middle of the design will go.
2. **Stack the layers:** outer hoop ring on the table, stabilizer on it, fabric on top (right side up).
3. **Line up the center cross** with the marks on the hoop edges (most hoops have notches at the top, bottom, and sides). Many hoops come with a clear plastic grid template that snaps in to help.
4. **Press the inner ring in** firmly, all the way around.

5. **Tighten the screw** and tug the fabric gently at the edges until it is drum-tight. Tap it: it should sound like a drum. Don't stretch it so far that the grain bends.
6. **Check the orientation.** The hoop's connector bracket goes toward the machine's arm, and the top of your design faces away from you. Check the arrow or "top" marking on the hoop.
7. **Attach the hoop:** with the presser foot **up**, slide the hoop's bracket into the arm's clamp until it clicks or locks. Then lower the presser foot.

Check: press the fabric in the center with a fingertip. It should barely give, and the fabric should not slide inside the rings.

Hand vs machine: on a hand hoop you re-tighten as you go. On the machine you cannot touch it once it starts, and a loose hoop is the most common cause of letters that look wavy or outlines that don't line up. Get it right now.

Step 5: Load the Henderson pattern

Goal: "Henderson" over "est. 2007" on the screen, centered in the hoop. There are two ways; do **Way A** first, then try **Way B** on a second run and compare.

Way A: use the ready-made file (recommended first)

The file `henderson-est-2007.dst` is in your Embroidery-Machine repository. It is bold serif lettering, 106 x 29 mm (about 4.2" x 1.1"), roughly 4,700 stitches. Navy for "Henderson", then a stop, then old gold for "est. 2007".



1. **Download** the `.dst` file to your computer.
2. **Copy it** to the root of a USB flash drive (not inside a folder).
3. **Plug the drive** into the machine's USB port.

4. On the touchscreen, open the **USB** (or memory) menu and tap the file. A preview appears.
5. **Confirm the hoop size** shows 5.5″ x 5.5″ (140 x 140 mm). If the screen asks, choose that hoop.
6. **Leave the size alone** for the first run. Resizing a stitch file by more than about 10–20% can make stitches too dense or too loose.

Hand vs machine: a .DST file is like a stitch chart that the machine reads itself. It holds every stitch position, but not the thread colors. The machine shows default colors on screen; the real colors are whatever spools you put on.

What you'll see on screen, and why it's normal:

- **Different colors than this guide.** A .DST file stores stitch positions, not thread colors. The machine fills in its own default colors (often red, blue, yellow) for each color block. Ignore them: what matters is the number of blocks and the order. Thread the spools this guide lists at each stop. Some machines let you tap a color block and pick a closer color, which is only for the preview.
- **Thin lines connecting letters and lines.** These are **jumps**: places where the machine ties off, trims the thread, lifts the needle, and moves to the next letter. The preview draws them so you can see the path; they are not stitched. If your machine leaves long threads across the fabric instead of trimming them, look in its settings for an automatic trim or jump-trim option and turn it on, or snip the threads after stitching.

Way B: type it on the touchscreen

The EOC07 has built-in fonts, so you can build the same design without a computer.

1. On the home screen, tap **Lettering** (it may be labelled Text, Font, or "ABC").
2. Pick a font. A bold block or serif font is easiest for a first try.
3. Type `Henderson`, then add a new line (look for a return or line-break key) and type `est. 2007`. If your screen only allows one line, make two text objects and drag "est. 2007" under "Henderson".
4. Set the size: about 15 mm tall letters for "Henderson" and 7–8 mm for "est. 2007". Letters smaller than 6 mm get hard to read in thread.
5. Tap **center** (or align) so the design sits in the middle of the hoop.
6. Save it to memory so you can use it again.

Try it: stitch Way A and Way B side by side on the same scrap. Compare how the fonts, sizes, and stitch types look. Built-in lettering often uses **satin stitch** (shiny side-to-side columns), while the file uses **fill stitch** (rows of short stitches, like tiny tatami weave).

Second design: Henderson with a four-leaf clover

Once the plain version stitches cleanly, try `henderson-est-2007-clover.dst` from the same folder. It is the same lettering with a small St. Patrick's Day four-leaf clover on each side of "est. 2007": 106 x 32 mm (about 4.2" x 1.3"), roughly 6,400 stitches, still in the 5.5" x 5.5" hoop.



It has **three colors**, so the machine stops twice:

1. **Forest green:** "Henderson".
2. **Kelly green:** both clovers.
3. **Old gold:** "est. 2007".

Load and run it exactly like Way A. At each stop, change only the top thread; the white bobbin stays in. For an all-green patch, keep the same spool and press Start/Stop at each stop.

Try it: watch the clover stitch. Each leaf fills in its own direction, so the four leaves catch the light differently and look separate even though they're one color.

Step 6: Run the machine

Goal: a finished stitch-out. Henderson takes roughly 6–8 minutes at full speed; stay at the machine the whole time for your first runs.

1. **Lower the presser foot** (if the machine doesn't do it automatically).

2. **Trace the design.** Find the **Trace**, **Frame**, or **Outline** button. The hoop moves around the outer edge of the design without stitching. Watch the needle: it must stay inside the hoop and never come near the plastic edge. If it gets close, move the design on screen and trace again.
3. **Hold the thread tail** lightly with one finger for the first 5–6 stitches so it doesn't get pulled under.
4. **Press Start/Stop.** Start at a slow speed (the speed setting is on the screen); you can raise it once you trust it.
5. **Watch the first 30 seconds.** Stitches should lie flat with no white showing on top. If you hear grinding or see loops forming, press Start/Stop right away.
6. **Snip jump threads** when the machine pauses, or leave them for the end. Never reach near the needle while it's running.
7. **Color change.** After "Henderson" the machine stops and shows the next color. Raise the presser foot, cut the navy thread near the spool, pull it out **from the needle end** (pulling from the spool end drags it backward through the tension discs), put on the gold spool, and re-thread (Step 3). Press Start/Stop. For a one-color patch, just press Start/Stop again.
8. **Finish.** The machine stops and tells you it is done. It trims the threads if the auto-cutter is on.

Check: turn the hoop over (still attached is fine). You should see mostly white bobbin thread, with a thin line of top color along the edges of each letter. That's a balanced stitch. On satin lettering (Way B), the white shows as a stripe down the middle third of each column.

Hand vs machine: you're the supervisor now, not the stitcher. Your job is to watch the thread, listen to the sound, and stop fast. A steady rhythmic hum is good; a clack, grind, or sudden change in sound means stop.

Step 7: Finish the piece

Goal: a clean patch with no loose threads and the stabilizer trimmed.

1. **Raise the presser foot** and unclip the hoop from the arm.
2. **Loosen the hoop screw** and pop the fabric out. Don't pull the fabric to get it out; push the inner ring through from behind.
3. **Trim jump threads** on the front close to the fabric. Curved snips work best.

4. **Trim the stabilizer.** Cut-away: trim to about 1/4 inch around the letters, rounding the corners. Tear-away: hold the stitches with one thumb and tear gently away from them, a little at a time.
5. **Press** from the back with a medium iron on a towel, so you don't flatten the stitches.
6. **Rethread with the dark color** and put a fresh scrap in the hoop for your next try.

Hand vs machine: cut-away stabilizer is never fully removed, like the fabric backing on a finished cross-stitch. Leaving it keeps the letters crisp through washing.

Troubleshooting and practice experiments

When something goes wrong, press Start/Stop, then work down this order: **re-thread the top thread → re-seat the bobbin → change the needle → check hooping**. Those four fix most problems.

What you see	Likely cause	Fix
A nest of tangled thread under the fabric ("bird's nest")	Top thread not in tension discs or take-up lever	Cut the nest free from below, re-thread with presser foot **up**
White bobbin thread showing on top	Top tension too tight, or bobbin in backwards	Re-thread; re-seat bobbin; then lower top tension one step
Loops of top thread on the front	Top tension too loose	Re-thread with foot up; raise top tension one step
Top thread keeps breaking or shredding	Old or wrong needle, thread caught on spool notch	Fresh 75/11 needle; turn spool so the notch faces right; slow down
Needle breaks	Hoop hit the needle, or bent needle	Power off, replace needle, re-trace the design
Fabric puckers around letters	Too little stabilizer or loose hoop	Add a second layer of stabilizer; re-hoop drum-tight
Letters out of line or outlines gap from fill	Fabric shifted in the hoop	Re-hoop tighter; use spray adhesive between fabric and stabilizer

What you see	Likely cause	Fix
Machine won't start	Presser foot up, hoop not seated, or wrong hoop chosen	Lower foot; re-click hoop; match hoop on screen

Practice experiments (one scrap each)

- [ ] **Stabilizer test:** stitch the pattern once with stabilizer and once without. See the pucker difference for yourself.
- [ ] **Tension test:** stitch once at normal tension, once with top tension two steps tighter. Flip both over and compare the backs.
- [ ] **Speed test:** run at the slowest and fastest speeds. Note which looks better on your fabric.
- [ ] **Font test:** Way A file vs Way B on-screen lettering, side by side.
- [ ] **Fabric test:** try a dish towel (add a water-soluble topper on top so stitches don't sink into the loops) and a T-shirt (use cut-away stabilizer).
- [ ] **Placement test:** stitch the patch on a real item, like a tote bag or towel.

Keep the machine happy

- Brush lint from the bobbin area every few projects (power off, remove the needle plate if your manual allows it).
- Change the needle every 8–10 hours of stitching, or sooner if thread starts shredding.
- Oil only where and when the manual says. Many newer machines need little or none.

Sources

- Poolin EOC07 product page (Poolin Store): 7.9″ x 15″ area, 4 hoops, 7″ touchscreen, USB and Wi-Fi transfer
- Best Buy EOC07-RE listing: DST file format, box contents
- Poolin EOC07 Amazon bundle: Class 15 (SA156) bobbins, cut-away stabilizer
- Pattern file and generator script

Machine-specific steps (bobbin direction, threading path, menu names) follow common single-needle machine design; check them against the printed quick guide in your box.